The ESG Premium in Corporate Bond Markets

University of Bologna

May 17, 2024

Ekaterina Nemchenko, Jelena Cesnokova, Sarp Derin Görgün, Shahriyar Majidi, Mohammad Kaleji

---

1 Introduction

This study aims to estimate the ESG premium (the difference between the yields of ESG-certified and conventional bonds) in corporate bond markets by analyzing non-financial, fixed-coupon corporate bonds issued in North America and the European Union since January 1, 2020, using data from Refinitiv Eikon. The focus is on determining whether ESG-labeled bonds, including green and other sustainability-focused types, enjoy a pricing premium over conventional bonds.

Empirical evidence on the ESG premium is mixed. Flammer, 2021 and Tang and Zhang, 2020 provide methodologies for analyzing the effects of ESG characteristics on bond pricing, which will guide our approach. Studies like Immel et al., 2022 and Zerbib, 2019 suggest that market differentiation exists based on ESG ratings, with governance structures playing a significant role.

By employing rigorous econometric methods, this research will contribute to understanding the drivers and existence of the ESG premium. The findings will inform investors and policymakers about the financial implications of ESG factors in sustainable finance, providing insights into the viability and impact of these investment products.

2 Literature Review

Green bonds align investment with environmental sustainability. This review synthesizes recent findings on green bond pricing, premiums, and the impact of external reviews and ESG ratings, relevant to our project on enhancing green bond credibility.

The green bond premium refers to yield differences between green and conventional bonds. Dorfleitner et al., 2021 found a negative green premium, implying that investors accept lower yields for environmentally beneficial projects. Similarly, Immel et al., 2022 reported a negative ESG-premium, influenced by ESG ratings, with higher ratings leading to narrower spreads.

External reviews, such as certifications and second-party opinions, enhance green bond credibility. Dorfleitner et al., 2021 found that these reviews lead to lower yields, indicating investor trust. Flammer, 2020 noted that third-party certifications improve both financial and environmental performance, reinforcing investor confidence. Moreover, ESG ratings assess a firm's environmental, social, and governance performance. Higher ESG ratings are linked to lower yield spreads, suggesting perceived lower risk and greater sustainability (Immel et al., 2022).

3 Data Pre-Processing and Summary Statistics

The data set subject to this analysis consists of conventional and ESG-labeled, still outstanding as of the day of the conduction of this analysis, corporate bonds issued by non-financial firms in Northern America and The European Union since the first quarter of the year 2020, compiled from the Refinitiv Eikon database on April 24, 2024. Perhaps the most important variable that almost the entire analysis relies on is the yield to maturity (computed at the extraction date), even though imperfect, nevertheless the single best gauge of the expected net rate of return of the bonds. For observations that miss this value are as good as useless to the analysis, the only observations that were extracted from the database were the ones with non-missing yield to maturities. Moreover, observations for which the ESG status, Maturity, and Amount Outstanding were missing or were equal to zero were also excluded. The data set contains a total of 6326 bonds, 492 of which are ESG certified, and the rest are conventional "brown" bonds. Some of the observations had extremely high values for yield to maturity and seemed to be outliers. As a first step to eliminate the potential adverse effect of these values on the analyses, following in the footsteps of the great Micheal Burry, one of the very few people who predicted the 2007-2008 financial crisis, these observations were manually examined. The firm with the highest yield to maturity was Modern Land (China) Co Ltd. This company was incorrectly listed in the Rifinitiv Eikon database as a US based company, whereas its headquarter is in Beijing, China. Therefore all bonds issued by this firm were removed from the data set. The firm with second highest yield bonds was an Indian based company named Future Retail Ltd, which was again incorrectly listed as a US based company. All other bonds with unreasonably high yield to maturities were also manually examined and removed, except for the pharmaceuticals (because it is in the nature of pharmaceutical projects to be very risky at times), and the ones that have reclaimed almost all of their debts. After removing these incorrectly listed and outlier observations, there remained 6305 bonds. The remaining observations will once more be tested for being outliers based on their leverage values in section 5.1. A summary of the cleaned data set is presented in figure 1, amount outstanding and amount issued for all bonds are denominated in US dollars. Ever since 2021, the total amount and the count of ESG bonds issued per year had declined. In panel b of figure 1, some features of the ESG bonds and conventional bonds in the sample are contrasted. The median bond in both sub samples has an outstanding amount of 500 million dollars, with the median conventional bond offering both a higher coupon rate and a higher yield to maturity compared to the median ESG bond. This might be interpreted as an early sign of a possibly negative ESG-premium, however, no conclusions can be drawn at this point without further investigation.

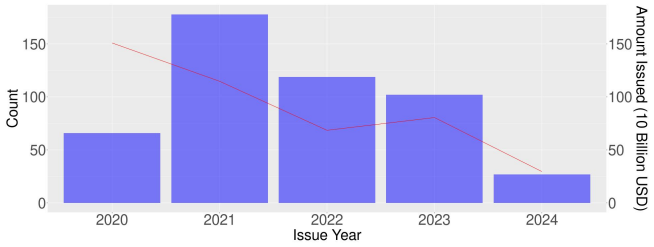
4 Methodology

The fundamental theorem of asset pricing states there exists a positive stochastic discount factor m that prices any stream of uncertain payoffs x:

$$p = \mathbb{E}[mx] \tag{1}$$

This fundamental equation is often written in terms of the covariance of the stochastic discount factor with the net rate of return of the asset:

$$\mathbb{E}[r] = r_f - cov(\eta, r) \tag{2}$$



(a) Total count (blue bars) and total amount (red line) of ESG bonds issued.

	Min	Median	Mean	Max
Coupon (ESG)	0.00	3.32	3.44	11.80
Coupon (Conventional)	0.00	4.23	4.30	14.50
Yield to Maturity (ESG)	2.99	5.42	5.59	27.83
Yield to Maturity (Conventional)	-3.22	5.76	6.48	198.98
Amount Outstanding - ESG (Millions of Dollars)	0.05	500	541	4262
Amount Outstanding - Conventional (Millions of Dollars)	0.001	500	648	31960
# Bonds	6305		# ESG Bonds	492

(b) Summary Statistics.

Figure 1: Summary Statistics.

Where $\mathbb{E}[r]$ is the expected net rate of return of the asset, r_f is the risk-free rate and $\eta \equiv (1 + r_f)m$ is the normalized stochastic discount factor. Since this equation holds for any type of asset (conventional, or ESG-labeled), the difference between the expected rates of returns of these two classes of assets is:

$$\mathbb{E}[r_{ESG}] - \mathbb{E}[r_{conv}] = -cov(\eta, r_{ESG}) + cov(\eta, r_{conv}) = cov(\eta, r_{conv} - r_{ESG}) \tag{3}$$

Where the last equality holds because the covariance is a linear operator, and the term on the left hand side is the so called ESG-premium. In order to proceed, an asset pricing model that connects the stochastic discount factor to data is required. Even though there are numerous models such as the consumption based-CAPM, ICAPM, and dozens of General Equilibrium models that serve this purpose, when it comes to empirical works such as this one, the ultimate choice is a statistical model.

Two such models are used in this analysis, a regression model, and a matching model. In the regression model, a linear model is defined to specify equation (2) in terms of the available variables. By controlling for fixed effects and including a dummy variable indicating the type of the bond (ESG or conventional) in the model, the existence of an ESG-premium could be tested by estimating the coefficient of the dummy variable. The matching analysis is built on the basis of equation (3). If for each ESG-certified bond, an almost identical conventional bond issued by the same firm could be found, then the covariances between the stochastic discount factor and yield to maturities of these bonds differ only to the extent that one is of the ESG type and the other is not. This difference is, by definition, the return adjustment for the bond being ESG-certified, the ESG-premium.

5 Regression Analysis

The regression model used in this analysis is defined as follows:

$$y_i = \beta_0 + \beta_1 \times ESGBond_i + \beta_2 \times Coupon_i + \beta_3 \times Maturity_i + \beta_4 \times InvestmentGrade_i + \tag{4}$$
$$\beta_5 \times HighYield_i + \beta_6 \times ln(AmountOutstanding_i) + Seniority\,FE + IssueYear\,FE + Issuer\,FE +$$

Where y_i is the yield to maturity calculated at the date of extraction of the data and ESGBond, the most important regressor in this analysis, is a dummy variable which equals one if the bond is ESG-certified (whether self-certified or certified by a recognized institution) and zero otherwise. A fully specified model requires the inclusion of variables that capture interest rate risk, credit risk and liquidity risk among others that affect the yield to maturity of the bonds. To gauge the interest rate risk, coupon rate and days to maturity (difference in days between the maturity date and data extraction date, April 24, 2024) were included in the model. For credit risk, exact credit ratings and probabilities of default or the financial ratios used for the assessment of credit risk of the firms were not available, the only available measurement was the general specification of the investment class of the bonds. InvestmentGrade is a dummy variable that equals one if the bond is classified as InvestmentGrade and zero otherwise. Likewise, HighYield is the dummy variable corresponding to low credit, HighYield bonds. All other bonds were treated as if though they were unrated. The natural logarithm of AmountOutstanding (instead of AmountIssued because yield to maturities were computed at the data extraction date and not at the issuance date) and Seniority fixed effects account for exposure at default risk. Finally, to control for yearly trends and firm specific factors, IssueYear fixed effects and Issuer fixed effects were included in the model. Issuer fixed effects include country of issue and sector. There were no plausible measurements (the Bid-Ask spread or volume for instance) available to assess the liquidity risk of the bonds. Even though it is fairly unlikely for liquidity risk to be correlated with the bond being ESG-certified, lack of variables that gauge the liquidity risk might potentially lead to biased estimates of the parameters of the model.

5.1 Results

At first, all observations were included in the regression. The average leverage value of the observations was 0.0127 However, 525 observations had leverage values higher than twice the average leverage value. These outliers were removed and the regression was carried out once more after having excluded the outliers. However, the results did not change significantly. The estimated parameters are reported in table 1.

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	3.1620	0.8070	3.92	0.0001
Coupon	0.8324	0.0413	20.15	0.0000
ln_Amount_Outstanding_USD	-0.0310	0.0260	-1.19	0.2326
ESG_Bond1	0.2127	0.2141	0.99	0.3205
Bond_GradeInvestment Grade	-0.7591	0.1789	-4.24	0.0000
Bond_GradeUnrated	2.3895	0.3228	7.40	0.0000
Days_to_Maturity	-0.0001	0.0000	-5.45	0.0000

(a)

	Estimate	Std. Error	t value	Pr(>|t|)
Issue_Year2021	0.3201	0.1385	2.31	0.0208
Issue_Year2022	-1.3804	0.1675	-8.24	0.0000
Issue_Year2023	-2.1675	0.1817	-11.93	0.0000
Issue_Year2024	-2.1478	0.2404	-8.93	0.0000
Measures of Goodness of Fit	F-Statistic		p-value	Adjusted R-squared
	29.16 (47,5732)DF		00.00	0.1864

(b)

Table 1: Regression Analysis Results.

The estimated ESG-premium from the regression model was 21.27 basis points, a relatively large positive premium, however, given the high p-value of 0.3205, it is statistically insignificant. Surprisingly, the effect of Days to Maturity on the Yield to Maturity, even though statistically significant, is practically insignificant. Therefore, the regression model might have been misspecified, and the estimates biased due to the omission of some important, unavailable variables.

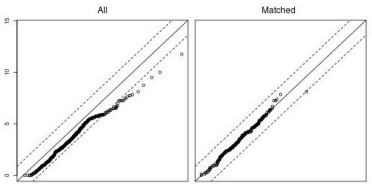
6 Matching Analysis

As an alternative approach to test for the existence of an ESG-premium, a matching methodology was used. The aim of this method is to estimate the average treatment effect for treated units in the population. The matching method is similar to that of Flammer, 2020. The treatment variable is the same dummy variable as in the regression model, ESGBond, and the covariates are Issuer, BondGrade (InvestmentGrade, HighYield, or Unrated), Seniority, Coupon, Maturity, ln(AmountOutstanding), and Issue Date. The first three covariates, Issuer, BondGrade, and Seniority are exactly matched to make sure that the pairs of bonds in the strata are issued by the same firm and have the same grade and the same exposure at default status. The rest of the covariates were matched using the nearest neighbor method with the Generalized Linear Model distance for the propensity score. The matching expression is defined as follows:

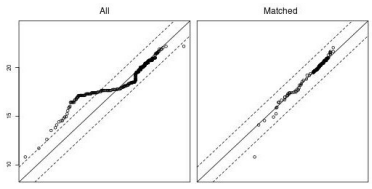
$$\begin{aligned} ESGBond_{ig} \sim & Issuer_{ig} + BondGrade_{ig} + Seniority_{ig} \\ & + Coupon_{ig} + Maturity_{ig} + ln(AmountOutstanding_{ig}) \end{aligned} \tag{5}$$

6.1 Results

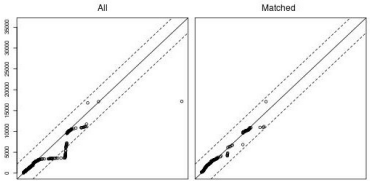
Out of the initial 6305 bonds, 492 were ESG bonds, and only 224 of them were matched with a bond in the control unit. Figure 2 shows the eQQ plots of each covariate. All covariates are fairly balanced after matching, indicating that the match was successful. The



(a) Coupon



(b) ln(Amount Outstanding)



(c) Maturity

Figure 2: Value of each covariate for the treated units (the y-axis) vs. value of each covariate in the control group (x-axis)

estimated difference between the mean yield to maturities of the ESG and conventional

bonds is reported in table 2. The estimate yielded a negative ESG premium of 9 basis points. However, given the standard error, this estimate is not significantly different from zero.

term	contrast	estimate	std.error	statistic	p.value	conf.low	conf.high
ESG_Bond	$\mathbb{E}[r_{ESG}] - \mathbb{E}[r_{conv}]$	-0.07	0.10	-0.65	0.51	-0.27	0.13
Sample Sizes			Matched		Unmatched		Discarded
# ESG Bonds	492		224		268		0
# Conventional Bonds	5813		224		5589		0

Table 2: Matching Analysis Results.

7 Discussion

Our regression analysis estimated a positive ESG premium of 21.27 basis points, which was statistically insignificant (p-value = 32.05%). This suggests that ESG bonds do not significantly differ in yield compared to conventional bonds in our sample. The fact that the estimated premium was positive is in contradiction to the other analysis carried out in this paper and the results of the other authors, Dorfleitner et al., 2021 and Immel et al., 2022 who have found a negative premium. We offer the explanation of a positive bias in the OLS estimators of the regression model due to the omission of variables that assess the credit and liquidity risks of the bonds.

Our matching analysis did not find a significant ESG premium either. The difference in mean yield to maturities between ESG and conventional bonds was in this case negative, and equal to 7 basis points, with again a very large p-value, implying lack of enough statistical evidence to support its significance. This finding is consistent with Hachenberg and Schiereck, 2018, who observed that while ESG ratings influence bond spreads, the effect is not universally significant.

The absence of a significant ESG premium in our study could be due to several factors. The period from 2020 onwards may represent a maturation phase in the ESG bond market, with initial premiums diminishing as the market stabilizes. Kapraun et al., 2021 noted that green bond premiums vary significantly based on issuer characteristics and external verifications, suggesting that perceived benefits of ESG bonds may now be fully integrated into their pricing.

Literature emphasizes the role of external reviews and ESG ratings in enhancing green bond credibility. Dorfleitner et al., 2021 and Flammer, 2020) found that certifications and third-party opinions reduce yields by increasing investor trust. In our study, variables such as coupon, maturity, and credit grade controlled for these factors, but results indicated that ESG certification alone does not significantly impact yields. This implies that the influence of ESG ratings and verifications may already be factored into the bond pricing by informed investors. Our findings also highlight the need for more nuanced models or additional variables to better capture market dynamics. The limited variables available, particularly for assessing liquidity risk, might have contributed to the limited explanatory power of our model. Zerbib, 2019 and other studies using linear regression models have faced similar challenges

8 Conclusion

This study is aimed to estimate the ESG premium in corporate bond markets by analyzing non-financial, fixed-coupon corporate bonds issued in North America and the European Union since January 1, 2020. Using data from Refinitiv Eikon, the study focused on determining whether ESG-labeled bonds, including green and other sustainability-focused types, enjoy a pricing premium over conventional bonds. The analysis utilized both regression and matching models to test for the presence of an ESG premium.

This research contributes to the literature on ESG investing by providing a comprehensive analysis of the ESG premium in corporate bond markets. It highlights the complexities and mixed results associated with ESG-labeled bonds, offering a detailed methodology for future research. For investors and policymakers, the findings suggest that the ESG premium is not statistically significant, indicating that ESG-labeled bonds may not necessarily offer a pricing advantage over conventional bonds. This insight can guide investment strategies and policy decisions regarding sustainable finance.

The study's limitations include the reliance on yield to maturity as the primary variable, potential model misspecifications, and the exclusion of proper credit and liquidity risk measurements. Additionally, the dataset's time frame and geographic focus may limit the generalizability of the results. Future research could expand the dataset to include a longer time frame and a broader geographic scope. Incorporating additional variables, such as exact credit ratings and liquidity risk measurements, could improve the robustness of the analysis. Further exploration of the impact of different ESG rating methodologies on bond pricing would also be valuable.

References

Dorfleitner, G., Utz, S., & Zhang, R. (2021). The pricing of green bonds: external reviews and the shades of green. *Review of Managerial Science*, 1–38.

Flammer, C. (2020). Green bonds: effectiveness and implications for public policy. *Environmental and Energy Policy and the Economy*, *1*(1), 95–128.

Flammer, C. (2021). Corporate green bonds. *Journal of financial economics*, *142*(2), 499–516.

Hachenberg, B., & Schiereck, D. (2018). Are green bonds priced differently from conventional bonds? *Journal of Asset Management*, *19*, 371–383.

Immel, M., Hachenberg, B., Kiesel, F., & Schiereck, D. (2022). Green bonds: Shades of green and brown. In *Risks related to environmental, social and governmental issues (ESG)* (pp. 21–34). Springer.

Kapraun, J., Latino, C., Scheins, C., & Schlag, C. (2021). (In)-credibly green: Which bonds trade at a green bond premium? *Proceedings of Paris December 2019 Finance Meeting EUROFIDAI-ESSEC*.

Tang, D. Y., & Zhang, Y. (2020). Do shareholders benefit from green bonds? *Journal of Corporate Finance*, *61*, 101427.

Zerbib, O. D. (2019). The effect of pro-environmental preferences on bond prices: Evidence from green bonds. *Journal of banking & finance*, *98*, 39–60.